

```
$ ./deploy --prod
```

Observabilidade 360°

Arquitetura e Orquestração de Microsserviços Resilientes

Aplicação-base: productivity-api · Spring Boot 3.5 · Java 21 · Docker · Helm · Kubernetes · Prometheus

CI/CD

Docker

Helm / K8s

Syslog-ng

Prom + Grafana

Containers vs VMs



Container

- Isolamento a nível de processo
- Compartilha o kernel do host
- Boot em segundos · footprint mínimo
- Imagem Alpine multi-stage ~250 MB



Máquina Virtual

- Virtualiza hardware completo
- Cada uma com seu próprio SO
- Boot em minutos · footprint alto
- Overhead de SO replicado por instância

~250 MB

imagem final (vs ~700 MB)

non-root

usuário appuser

multi-stage

JDK fica só no builder

segundos

tempo de boot

Como os recursos interagem no cluster

namespace: productivity-dev

Service

ClusterIP : 80

Deployment → Pod

startup · liveness · readiness

ConfigMap

DB_URL · profile

Secret

postgres-credentials

1 chart · 3 ambientes

dev sem HPA/Ingress · staging e prod ligam HPA, Ingress e mais réplicas — só os values-*.yaml mudam.

```
joaovictor@pop-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI$ kubectl get all -n productivity-dev
NAME                                READY   STATUS    RESTARTS   AGE
pod/postgres-0                      1/1     Running   0           33m
pod/productivity-productivity-api-6866bd5cdb-wn9zm  1/1     Running   0           31m

NAME                                TYPE          CLUSTER-IP   EXTERNAL-IP   PORT(S)    AGE
service/postgres                    ClusterIP      None         <none>        5432/TCP   33m
service/productivity-productivity-api  ClusterIP    10.96.243.242 <none>        80/TCP     31m

NAME                                READY   UP-TO-DATE   AVAILABLE   AGE
deployment.apps/productivity-productivity-api  1/1     1             1           31m

NAME                                DESIRED   CURRENT   READY   AGE
replicaset.apps/productivity-productivity-api-6866bd5cdb  1         1         1       31m

NAME                                READY   AGE
statefulset.apps/postgres            1/1     33m
joaovictor@pop-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI$ kubectl get hpa,ingress -n productivity-dev
No resources found in productivity-dev namespace.
```

O "pulo do gato": rollback ao vivo



Ato 1 · Deploy bom

`helm list · health UP`



Ato 2 · Quebra proposital

`image.tag inexistente →
ImagePullBackOff`



Ato 3 · Rollback

`helm rollback → recuperação`

Resultado medido

MTTR 1 min 29 s

Deploy quebrado às 21:45:39 → rollback OK às 21:47:08.
O pod velho continuou servindo durante todo o ImagePullBackOff.

Zero requisições perdidas

Desastre controlado e recuperação

```
oavictor@pop-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI$ helm upgrade productivity ./productivity-api/helm/productivity-api \
-f productivity-api/helm/productivity-api/values.yaml \
-f productivity-api/helm/productivity-api/values-dev.yaml \
--set image.tag=versao-que-nao-existe \
--namespace productivity-dev
release "productivity" has been upgraded. Happy Helming!
NAME: productivity
LAST DEPLOYED: Mon Jun  8 22:00:27 2026
NAMESPACE: productivity-dev
STATUS: deployed
REVISION: 5
TEST SUITE: None
oavictor@pop-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI$ kubectl get pods -n productivity-dev -w
```

NAME	READY	STATUS	RESTARTS	AGE
postgres-0	1/1	Running	0	50m
productivity-productivity-api-6866bd5cdb-wn9zm	1/1	Running	0	48m
productivity-productivity-api-866697cb95-mj8t7	0/1	ErrImagePull	0	11s
productivity-productivity-api-866697cb95-mj8t7	0/1	ImagePullBackOff	0	15s

kubectl get pods -n productivity-dev -w

pod novo: ImagePullBackOff · pod velho: Running

Desastre controlado e recuperação

```
• ^Cjoaovictor@pop-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI$ helm rollback productivity 2 -n productivity-dev
Rollback was a success! Happy Helming!
• joaovictor@pop-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI$ kubectl get pods -n productivity-dev
NAME                                READY   STATUS    RESTARTS   AGE
postgres-0                          1/1     Running   0           52m
productivity-productivity-api-6866bd5cdb-wn9zm  1/1     Running   0           50m
• joaovictor@pop-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI$ helm history productivity -n productivity-dev
REVISION      UPDATED              STATUS          CHART              APP VERSION      DESCRIPTION
1             Mon Jun 8 21:11:43 2026    superseded       productivity-api-0.1.0  0.0.1            Install complete
2             Mon Jun 8 21:13:56 2026    superseded       productivity-api-0.1.0  0.0.1            Upgrade complete
3             Mon Jun 8 21:47:45 2026    superseded       productivity-api-0.1.0  0.0.1            Upgrade complete
4             Mon Jun 8 21:49:47 2026    superseded       productivity-api-0.1.0  0.0.1            Rollback to 2
5             Mon Jun 8 22:00:27 2026    superseded       productivity-api-0.1.0  0.0.1            Upgrade complete
6             Mon Jun 8 22:02:13 2026    deployed        productivity-api-0.1.0  0.0.1            Rollback to 2
❖ joaovictor@pop-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI$
```

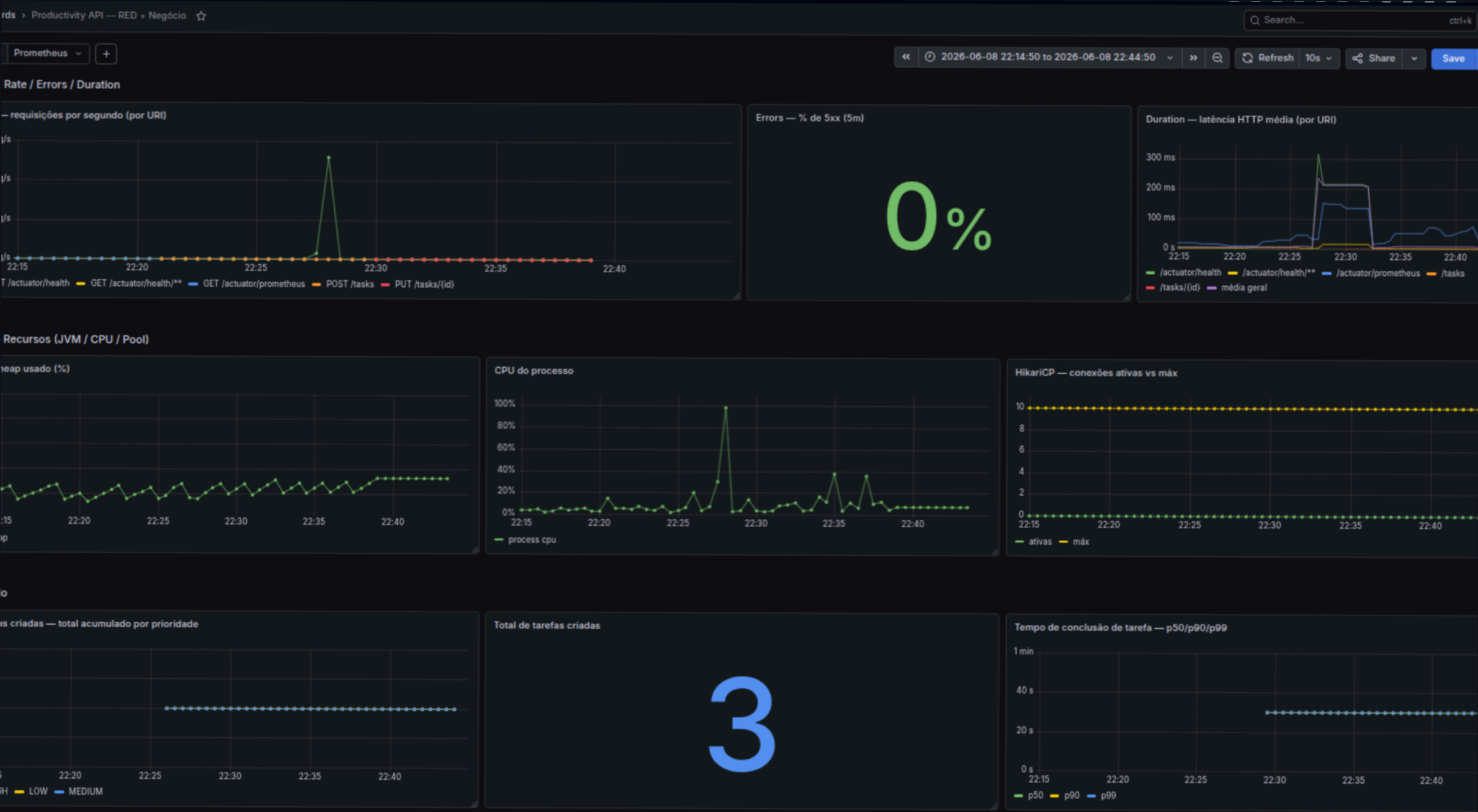
helm history productivity -n productivity-dev
(antes e depois do rollback)

Painel em tempo real — RED + USE + Negócio



RED

rate · % erro 5xx · latência



USE

heap JVM · CPU · HikariCP



Negócio

tarefas por prioridade

Scrape ativo e alertas baseados em SLO

Targets	
Query	Alerts
Status > Target health	
container="prometheus" endpoint="http-web" instance="10.244.0.13:9090" job="kube-prom-stack-kube-prometheus" namespace="observability" pod="prometheus-kube-prom-stack-kube-prometheus-0" service="kube-prom-stack-kube-prometheus" 29.104s ago 49ms	
e-prom-stack-kube-prometheus/1	
Labels	Last scrape
container="config-reloader" endpoint="reloader-web" instance="10.244.0.13:8080" job="kube-prom-stack-kube-prometheus" namespace="observability" pod="prometheus-kube-prom-stack-kube-prometheus-0" service="kube-prom-stack-kube-prometheus" 22.099s ago	
e-prom-stack-kube-state-metrics/0	
Labels	Last scrape
container="kube-state-metrics" endpoint="http" instance="10.244.0.10:8080" job="kube-state-metrics" namespace="observability" pod="kube-prom-stack-kube-state-metrics-6d5b77fb74-zwnwd" service="kube-prom-stack-kube-state-metrics" 21.664s ago	
e-prom-stack-prometheus-node-exporter/0	
Labels	Last scrape
container="node-exporter" endpoint="http-metrics" instance="172.18.0.2:9100" job="node-exporter" namespace="observability" pod="kube-prom-stack-prometheus-node-exporter-8pd4h" service="kube-prom-stack-prometheus-node-exporter" 12.263s ago 388ms	
productivity-productivity-api/0	
Labels	Last scrape
container="productivity-api" endpoint="http" instance="10.244.0.7:8080" job="productivity-productivity-api" namespace="productivity-dev" pod="productivity-productivity-api-6866bd5cdb-wn9zm" service="productivity-productivity-api" 2.474s ago	

5 alertas em prometheus-rules.yaml

- **HighErrorRate**
5xx > 1% por 5m
- **HighLatencyAvg**
latência média > 500ms por 10m
- **PodCrashLooping**
restarts > 0 em 15m
- **HikariPoolNearExhaustion**
conexões ativas/máx > 90%
- **ProductivityApiDown**
up == 0 por 2m

A caixa-preta que sobrevive ao container

"Um sistema sem logging adequado é como dirigir um carro sem painel." — Prof. Cloves Rocha

```
ip-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI/syslog-ng$ docker exec syslog-ng find /var/log/docker -type f
/var/log/docker/productivity-api/2026/06/09.log
/var/log/docker/productivity-api/2026/06/03.log
/var/log/docker/_errors/2026-06-09.log
/var/log/docker/_errors/2026-06-03.log
/var/log/docker/productivity-postgres/2026/06/09.log
/var/log/docker/productivity-postgres/2026/06/03.log
ip-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI/syslog-ng$ docker exec syslog-ng tail -n 15 /var/log/docker/productivity-api/2026/06/09.log
27 172.19.0.1 productivity-api[229]: ( ( ) \_ | ' | ' | | ' \_ | \ \ \ \
27 172.19.0.1 productivity-api[229]: \ \ / \_ ) | | ) | | | | | | ( | ) ) ) )
27 172.19.0.1 productivity-api[229]: ' | \_ | . \_ | | \_ | \_ , | / / / /
27 172.19.0.1 productivity-api[229]: =====|_|=====| \_ / = / / / /
27 172.19.0.1 productivity-api[229]: :: Spring Boot :: (v3.5.14)
27 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:27 INFO c.g.j.p.ProductivityApiApplication - Starting ProductivityApiApplication v0.0.1-SNAPSHOT using Java 21.0
- started by appuser in /app)
27 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:27 INFO c.g.j.p.ProductivityApiApplication - The following 1 profile is active: "prod"
38 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:38 INFO org.flywaydb.core.FlywayExecutor - Database: jdbc:postgresql://postgres:5432/productivity (PostgreSQL 16
38 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:38 INFO o.f.core.internal.command.DbValidate - Successfully validated 2 migrations (execution time 00:00.082s)
38 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:38 INFO o.f.core.internal.command.DbMigrate - Current version of schema "public": 2
38 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:38 INFO o.f.core.internal.command.DbMigrate - Schema "public" is up to date. No migration necessary.
42 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:42 WARN o.s.b.a.o.j.JpaBaseConfiguration$JpaWebConfiguration - spring.jpa.open-in-view is enabled by default. The
y be performed during view rendering. Explicitly configure spring.jpa.open-in-view to disable this warning
44 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:44 INFO c.g.j.p.ProductivityApiApplication - Started ProductivityApiApplication in 19.969 seconds (process runni
44 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:44 WARN o.s.c.events.SpringDocAppInitializer - SpringDoc /v3/api-docs endpoint is enabled by default. To disable
property 'springdoc.api-docs.enabled=false'
44 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:44 WARN o.s.c.events.SpringDocAppInitializer - SpringDoc /swagger-ui.html endpoint is enabled by default. To disa
the property 'springdoc.swagger-ui.enabled=false'
```

A caixa-preta que sobrevive ao container

"Um sistema sem logging adequado é como dirigir um carro sem painel." — Prof. Cloves Rocha

```
tor@pop-os:~/Documentos/Programação/Projetos de Estudo/ProductivityAPI/syslog-ng$ # 1. logs frescos na API de logging (8081)
n $(seq 1 5); do
-s -X POST http://localhost:8081/tasks \
"Content-Type: application/json" \
"{\"title\": \"Log test $i\", \"description\": \"gerando log\", \"priority\": \"LOW\", \"status\": \"PENDING\"}" >/dev/null
done

# 2. Parando o container da API
compose stop api

# 3. Tirando foto do container saindo
foto: container Exited, log ainda vivo

# 4. Verificando o status dos containers
compose ps
exec syslog-ng tail -n 8 /var/log/docker/productivity-api/2026/06/09.log
p 1/1
Container productivity-api-logging Stopped
```

	IMAGE	COMMAND	SERVICE	CREATED	STATUS	PORTS
productivity-postgres-logging	postgres:16-alpine	"docker-entrypoint.s..."	postgres	About an hour ago	Up About an hour	5432/tcp
syslog-ng	lscr.io/linuxserver/syslog-ng:latest	"/init"	syslog-ng	About an hour ago	Up About an hour	127.0.0.1:1514->5514/tcp, 127.0.0.1:1514->5514/tcp

```
00:25:38 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:38 INFO org.flywaydb.core.FlywayExecutor - Database: jdbc:postgresql://postgres:5432/productivity (PostgreSQL 16.14)
00:25:38 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:38 INFO o.f.core.internal.command.DbValidate - Successfully validated 2 migrations (execution time 00:00.082s)
00:25:38 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:38 INFO o.f.core.internal.command.DbMigrate - Current version of schema "public": 2
00:25:38 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:38 INFO o.f.core.internal.command.DbMigrate - Schema "public" is up to date. No migration necessary.
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00:25:44 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:44 INFO c.g.j.p.ProductivityApiApplication - Started ProductivityApiApplication in 19.969 seconds (process running for
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the property 'springdoc.api-docs.enabled=false'
00:25:44 172.19.0.1 productivity-api[229]: 2026-06-09 00:25:44 WARN o.s.c.events.SpringDocAppInitializer - SpringDoc /swagger-ui.html endpoint is enabled by default. To disable it
set the property 'springdoc.swagger-ui.enabled=false'
```

O pipeline bloqueia falhas antes do deploy



All checks have passed

6 successful checks

✓	CD / Build & Push Image (push)	Successful in 2m	Details
✓	CI / Build, Test & Coverage (push)	Successful in 59s	Details
✓	CD / Deploy to Production (push)	Successful in 3s	Details
✓	CD / Deploy to Staging (push)	Successful in 3s	Details
✓	CI / Lint (Spotless) (push)	Successful in 12s	Details
✓	CI / Security Scan (OWASP) (push)	Successful in 54s	Details

Tudo versionado, reproduzível do zero



CI/CD

ci.yml + cd.yml · GHCR



Containers

Dockerfile multi-stage · compose



Helm Chart

3 values · HPA · Ingress · probes



Logging

Syslog-ng · hierarquia por data



Métricas

Prometheus · Grafana · 5 alertas



Qualidade

324 testes · JaCoCo · SpotBugs

Entregue: README · docker-compose.yml · pasta do Helm Chart · *slides (este PDF)*

EQUUIPE



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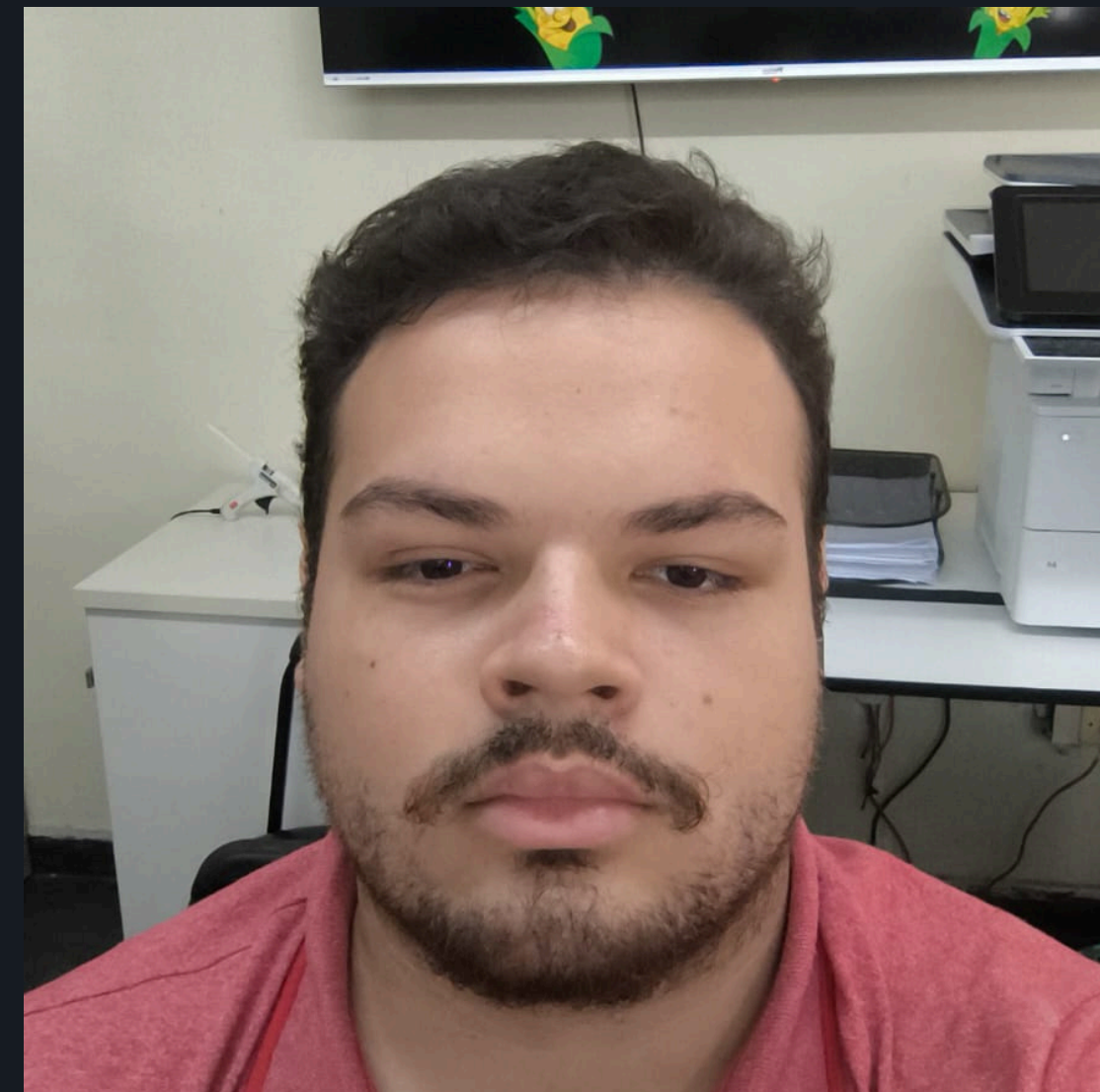
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